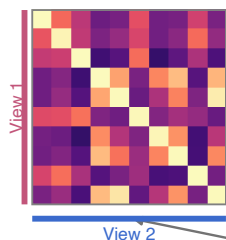


a

both objectives act on the same CLS representation

$$\mathcal{L} = \mathcal{L}_{\text{recon}} + 0.05 \mathcal{L}_{\text{InfoNCE}}$$

InfoNCE
cross-view
similarity matrix



reconstruction
of CpGs withheld
from each view

Decoder

predicted at withheld CpGs

CLS₁

⋮

CLS_B

CLS
representations

CLS₁

⋮

CLS_B

Shared MethyLLama encoder

6 transformer blocks · 256-d · genomic-rank RoPE

View 1

View 2

profiles
restricted
to each
subset

subset 1

subset 2

two independent
50% CpG subsets

Pretraining corpus

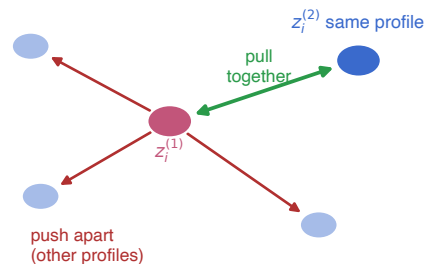
169,120 profiles × 49,156 CpGs

profiles

CpG sites

b

Contrastive objective



c

One transformer block

CLS output = sample representation

Add ⊕ residual

SwiGLU feed-forward

RMSNorm

Add ⊕ residual

Multi-head self-attention
+ genomic-rank RoPE

RMSNorm

token sequence in

× 6

d

Token construction



CpGs ordered by genomic rank

CpG identity
embedding

β-value
encoder

each token carries both which CpG and how methylated